universe would have been high enough to produce these particles copiously. In general one would not expect them to survive to the present day, as such super-heavy particles are usually unstable, relinquishing their enormous mass by decaying into a cascade of ever lighter particles, ending with the familiar, relatively light particles in the world around us. However, it is possible that such a super-heavy vibrational string state—a relic from the big bang—did survive to the present. Finding such particles, as we discuss more fully in Chapter 9, would be a monumental discovery, to say the least.

Gravity and Quantum Mechanics in String Theory

The unified framework that string theory presents is compelling. But its real attraction is the ability to ameliorate the hostilities between the gravitational force and quantum mechanics. Recall that the problem in merging general relativity and quantum mechanics turns up when the central tenet of the former—that space and time constitute a smoothly curving geometrical structure—confronts the essential feature of the latter—that everything in the universe, including the fabric of space and time, undergoes quantum fluctuations that become increasingly turbulent when probed on smaller and smaller distance scales. On sub-Planck-scale distances, the quantum undulations are so violent that they destroy the notion of a smoothly curving geometrical space; this means that general relativity breaks down.

String theory softens the violent quantum undulations by "smearing" out the short-distance properties of space. There is a rough and a more precise answer to the question of what this really means and how it resolves the conflict. We discuss each in turn.

The Rough Answer

Although it sounds unsophisticated, one way that we learn about the structure of an object is by hurling other things at it and observing the precise way in which they are deflected. We are able to see things, for example, because our eyes collect and our brains decode information carried by photons as they bounce off of objects being viewed. Particle accelerators are based on the same principle: They hurl bits of matter such as electrons and protons at each other as well as at other targets, and elaborate detectors analyze the resulting spray of debris to determine the architecture of the objects involved.